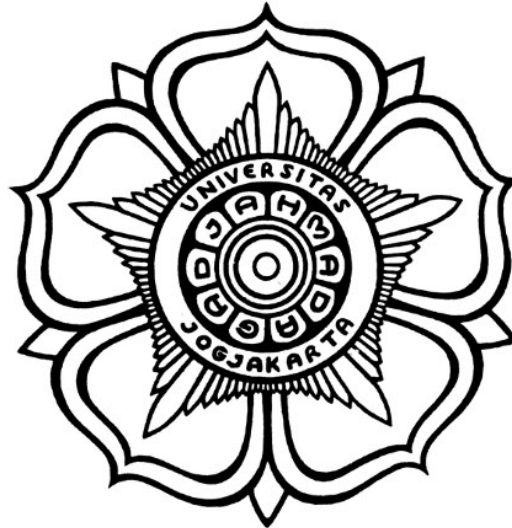


**VEHICLE-LIKELIHOOD SPATIOTEMPORAL BASELINE
ANALYSIS OF RING ROAD UTARA USING ACTIVE MOBILE
POSITIONING DATA**

THESIS



THE SUSTAINABLE DEVELOPMENT GOALS
Industry, Innovation and Infrastructure
Affordable and Clean Energy
Climate Action

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YOGYAKARTA
«Defense Year»**

ENDORSEMENT PAGE

VEHICLE-LIKELIHOOD SPATIOTEMPORAL BASELINE ANALYSIS OF RING ROAD UTARA USING ACTIVE MOBILE POSITIONING DATA

THESIS

Proposed as A Requirement to Obtain
Undergraduate Degree (*Sarjana Teknik*)
in Department of Electrical Engineering and Information Technology
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PAGE OF DEDICATION

This thesis is dedicated to the people who have supported the author's academic journey.

PREFACE

Praise be to Allah SWT for His mercy, blessings, and guidance, through which this undergraduate thesis has been completed. In preparing this thesis, the author received guidance, assistance, and support from many parties. Therefore, the author would like to express sincere gratitude to:

1. Person 1, for the guidance, support, and encouragement provided throughout the completion of this thesis.
2. Person 2, for the assistance, advice, and constructive feedback given during the research and writing process.
3. <insert the names of other individuals or institutions to be acknowledged>

The author hopes that this thesis will provide benefits for readers and contribute to future research and practice in transportation-data analysis.

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NOMENCLATURE AND ABBREVIATION

MPD	=	Mobile Positioning Data
MAID	=	Mobile Advertising Identifier
RRU	=	Ring Road Utara
OD	=	Origin-Destination
GPS	=	Global Positioning System
HMM	=	Hidden Markov Model
PPKM	=	Public Activity Restrictions Enforcement (<i>Pemberlakuan Pembatasan Kegiatan Masyarakat</i>)
WIB	=	Western Indonesian Time
UGM	=	Universitas Gadjah Mada
DIY	=	Special Region of Yogyakarta (<i>Daerah Istimewa Yogyakarta</i>)
<i>fishbone</i>	=	Road network representation that includes the main corridor and connecting roads around intersections
<i>backbone</i>	=	Main-corridor road network representation used to filter movements along Ring Road Utara
<i>catchment area</i>	=	Spatial area from which users of a corridor or intersection are estimated to originate
<i>anchor point</i>	=	Estimated representative location frequently associated with a user's activity, such as a home location

INTISARI

The construction of the Yogyakarta–Solo toll road is expected to substantially alter traffic patterns along the Ring Road Utara (RRU) corridor; however, a quantitative pre-toll traffic baseline is not yet available. This study develops a vehicle-likelihood spatiotemporal baseline analysis for 6 signalized intersections along Ring Road Utara using large-scale active Mobile Positioning Data (MPD). The analysis dataset consists of 312,935 GPS points from 8,980 unique devices classified as vehicle-likelihood MAIDs on the RRU *fishbone* network, all of which were retained by the current QGIS-curated *backbone* filter for OD and *catchment area* analyses during the period from October 2021 to June 2022. The methodology includes speed-feature-based transport-mode classification, geometric *map matching*, time-gap-based trajectory segmentation using a 10-minute threshold, and analyses of *origin-destination* (OD) matrices, traffic density, *catchment area*, and *turning movement*. The results show that Condongcatur is the densest intersection, with an average of 49.4 MAIDs/day, and that the highest densities generally occur from midday to late afternoon. The *backbone*-based OD matrix identifies 33,288 trips, with weekday flows dominating at 69.1%. *Catchment area* estimation using the *anchor point* model indicates that most RRU users reside near the corridor. *Turning movement* analysis is treated as exploratory because many trips are observed in only one intersection zone. This study provides a pre-toll spatiotemporal traffic baseline that can support future toll-road impact assessment.

Keywords: Mobile Positioning Data, vehicle-likelihood, spatiotemporal baseline, origin-destination, Ring Road Utara, pre-toll traffic

ABSTRACT

The construction of the Yogyakarta–Solo toll road is expected to substantially alter traffic patterns along the Ring Road Utara (RRU) corridor; however, a quantitative pre-toll traffic baseline is not yet available. This study develops a vehicle-likelihood spatiotemporal baseline analysis for 6 signalized intersections along Ring Road Utara using large-scale active Mobile Positioning Data (MPD). The analysis dataset consists of 312,935 GPS points from 8,980 unique devices classified as vehicle-likelihood MAIDs on the RRU fishbone network, all of which were retained by the current QGIS-curated backbone filter for OD and catchment area analyses during the period from October 2021 to June 2022. The methodology includes speed-feature-based transport-mode classification, geometric map matching, time-gap-based trajectory segmentation using a 10-minute threshold, and analyses of origin-destination (OD) matrices, traffic density, catchment area, and turning movement. The results show that Condongcatur is the densest intersection, with an average of 49.4 MAIDs/day, and that the highest densities generally occur from midday to late afternoon. The backbone-based OD matrix identifies 33,288 trips, with weekday flows dominating at 69.1%. Catchment area estimation using the anchor point model indicates that most RRU users reside near the corridor. Turning movement analysis is treated as exploratory because many trips are observed in only one intersection zone. This study provides a pre-toll spatiotemporal traffic baseline that can support future toll-road impact assessment.

Keywords: Mobile Positioning Data, vehicle-likelihood, spatiotemporal baseline, origin-destination, Ring Road Utara, pre-toll traffic

CHAPTER I

INTRODUCTION

1.1 Background

The City of Yogyakarta and its surrounding areas constitute one of Indonesia's urban regions that continues to experience growth in population and economic activity. This growth directly affects the increasing volume of traffic, particularly on major arterial routes that connect various areas within the Special Region of Yogyakarta (DIY). One arterial corridor that plays an important role in the regional transportation system is Ring Road Utara (RRU), which extends for approximately 9 km from Kronggahan Intersection in the west to UPN Intersection in the east.

Ring Road Utara connects 6 major signalized intersections, namely Kronggahan, Jombor, Monjali, Kentungan, Condongcatur, and UPN. This corridor not only functions as a connector between areas in Sleman Regency, but also provides access to major educational institutions such as Universitas Gadjah Mada, Universitas Negeri Yogyakarta, and Universitas Pembangunan Nasional, as well as to dense residential and commercial areas.

The ongoing construction of the Yogyakarta–Solo toll road will build an *elevated toll road* directly above the Ring Road Utara corridor. Four *on/off ramp* locations are planned along this corridor, which will fundamentally alter surface traffic-flow patterns. To evaluate the impacts of the construction and operation of the toll road, *baseline* data on traffic conditions prior to construction are required.

Conventional traffic data collection, such as manual intersection surveys or the use of *loop detectors*, has limitations in terms of spatial coverage, observation duration, and implementation cost [1]. Conversely, advances in *mobile positioning* technology have opened new opportunities for large-scale mobility data collection. Active *Mobile Positioning Data* (MPD), obtained from GPS devices on *smartphones* with user consent, offers substantially broader spatial and temporal coverage than conventional methods [2].

The active MPD dataset available in this study covers GPS data from October 2021 to June 2022, a period before toll-road construction began along the Ring Road Utara corridor. The working dataset at the initial stage of the study-area pipeline consists of tens of millions of GPS *pings* from hundreds of thousands of unique devices (*Mobile Advertising ID / MAID*), before being filtered to retain only data located on the Ring Road Utara network. These data provide a unique opportunity to develop a comprehensive traffic *baseline* characterization before major infrastructure changes occur.

Previous research by Nurlita (2024) demonstrated that active MPD can be used to

identify speed and traffic-flow patterns on Jalan Malioboro, although with limitations due to the low data sampling frequency [3]. Other research within the same research group has developed methods for classifying vehicle and non-vehicle movements using GPS data at intersections [4]. However, no study has specifically characterized intersection-level traffic along the Ring Road Utara corridor using active MPD.

1.2 Problem Formulation

Based on the background described above, the research problems in this study are formulated as follows:

1. What are the distribution patterns of *origin-destination* (OD) trips at the 6 major intersections on Ring Road Utara, and how do these patterns vary by time and day?
2. What are the spatiotemporal patterns of *traffic density* at each intersection, and which intersections experience the highest density during peak hours?
3. What is the distribution of the origin areas (*catchment area*) of vehicle users traveling through Ring Road Utara, and does this distribution differ across intersections?
4. What dominant *turning movement* patterns occur at each signalized intersection?

1.3 Research Objectives

This study aims to:

1. Extract and analyze *origin-destination* (OD) matrices at 6 intersections on Ring Road Utara using active MPD.
2. Estimate and visualize spatiotemporal traffic-density patterns at each intersection.
3. Identify the *catchment area* of Ring Road Utara users based on estimated home locations.
4. Classify *turning movement* patterns at each signalized intersection.

1.4 Research Scope and Limitations

This study has several limitations, as follows:

1. **Research object:** Analysis of traffic characteristics at 6 signalized intersections on Ring Road Utara, Sleman Regency, DIY.
2. **Data source:** Active MPD obtained through collaboration between the research team and a telecommunications company in Indonesia.
3. **Data period:** October 2021 to June 2022.

4. **Map matching method:** Geometric matching (*nearest-edge*) based on the taxonomy of Quddus et al. (2007), rather than a *Hidden Markov Model*.
5. **Road network:** A *fishbone* road-network model for intersection preprocessing and a *backbone-only* model that preserves the main west–east RRU route for OD and *catchment area* analyses.
6. This study analyzes *smartphone* data that require user consent and therefore does not represent the entire population of road users.
7. The low GPS sampling frequency limits the accuracy of the *turning movement* analysis.

1.5 Research Contributions

The results of this study are expected to provide benefits for:

1. **Transportation policymakers**, by providing *baseline* data on pre-toll traffic conditions that can be used to evaluate the impact of the Yogyakarta–Solo toll-road construction on traffic along Ring Road Utara.
2. **Traffic planners**, in designing traffic-management scenarios at intersections affected by the presence of toll-road *on/off ramps*.
3. **Transportation researchers**, as a reference for the use of active MPD in large-scale traffic analysis in Indonesia, particularly in the context of infrastructure-impact evaluation.

1.6 Thesis Organization

Chapter I presents the introduction, consisting of the background, problem formulation, research objectives, research scope and limitations, and research contributions.

Chapter II presents the literature review and theoretical foundations, covering related literature and the fundamental concepts used in this study.

Chapter III presents the research methodology, including the research workflow, system architecture, and analytical techniques used.

Chapter IV presents the results and discussion, including the findings from the data analysis and discussion of those results.

Chapter V presents the conclusions and recommendations based on the results of the study.

CHAPTER II

LITERATURE REVIEW AND THEORETICAL FOUNDATIONS

2.1 Literature Review

Several previous studies relevant to this research are summarized in Table 2.1.

Table 2.1. Summary of Literature Review

Author (Year)	Method	Data	Main Findings
Nurlita (2024) [3]	<i>Path reconstruction</i> (HMM + Viterbi), temporal interpolation	Active MPD (GPS), Jl. Malioboro, Jan–Feb 2022	Identification of speed patterns (average 14.5 km/h) and traffic flow. Pearson correlation of 0.52 with Google Maps.
Alexander et al. (2015) [5]	<i>Trip extraction, purpose classification, expansion factor</i>	CDR (<i>Call Detail Records</i>), Boston, MA	OD matrices were highly correlated with household surveys at the city-level aggregation scale.
Calabrese et al. (2013) [2]	Mobile-phone <i>trace</i> analysis, validation against odometer data	Passive MPD, 1 million users, Boston	Mobile-phone data provide an adequate proxy for individual mobility.
Newson & Krumm (2009) [6]	<i>Hidden Markov Model</i> for <i>map matching</i>	GPS trajectories, Seattle	HMM accuracy decreases significantly at sampling intervals >30 seconds.
Quddus et al. (2007) [1]	Survey of <i>map matching</i> algorithms: geometric, topological, probabilistic	Literature review	Classification into 4 categories. Geometric methods are sufficient for simple road networks.
Zheng (2015) [7]	Survey of <i>trajectory data mining</i> : segmentation, classification, <i>clustering</i>	Literature review	Comprehensive <i>framework</i> . Segmentation thresholds are heuristic and application-dependent.
Zheng et al. (2008) [8] and Reddy et al. (2010) [9]	GPS feature-based transportation-mode classification using speed, acceleration, and speed percentiles	GPS trajectories from mobile devices	Speed features effectively distinguish walking, cycling, and motorized vehicles at the trajectory level.
Ahas et al. (2010) [10]	<i>Anchor point</i> model for estimating home and workplace locations	Passive MPD, Estonia	Validation against population-registration data showed high accuracy at the city level.
Barbosa et al. (2018) [11]	Survey of human-mobility models	Literature review	Comprehensive framework for mobility patterns using <i>big data</i> .

Based on the literature review above, several research gaps can be identified as

the motivation for this study:

1. Previous research within this research group (Nurlita, 2024) focused on the Jalan Malioboro corridor, a one-way tourist street, rather than a multi-intersection arterial corridor.
2. No study has characterized traffic at the intersection level along the Ring Road Utara corridor using active MPD.
3. The use of active MPD for OD matrix analysis, *catchment area* estimation, and *turning movement* classification at signalized intersections in Indonesia remains very limited.

2.2 Theoretical Foundations

2.2.1 Mobile Positioning Data (MPD)

Mobile Positioning Data (MPD) are data related to mobile-phone use that include location information from mobile-phone users collected by cellular network operators [2]. MPD can be divided into two main types:

1. **Active MPD:** Obtained through direct requests to mobile devices using GPS technology. It requires user permission and provides high accuracy (on the order of meters).
2. **Passive MPD:** Collected automatically by cellular operators without user interaction through *Call Detail Records* (CDR). It has low accuracy (100 meters in dense areas and up to 30 km in remote areas).

The dataset used in this study is active MPD with four feature columns, as shown in Table 2.2.

Table 2.2. Active MPD Feature Columns

Feature	Description
maid	Unique device identification number (<i>Mobile Advertising ID</i>)
latitude	Latitude coordinate of the device location
longitude	Longitude coordinate of the device location
timestamp	Sampling time in <i>Unix Timestamp</i> format

2.2.2 GPS-Based Transportation-Mode Classification

Transportation-mode classification from GPS data generally uses derived movement features such as maximum speed, average speed, acceleration, and speed percentiles. Zheng et al. [8] showed that speed and movement-change features can be used to

understand mobility and distinguish travel patterns. Reddy et al. [9] also used speed and acceleration features from mobile phones to distinguish walking, cycling, and motorized-vehicle modes. Because the dataset in this study does not include mode labels obtained from field surveys, classification is performed using transparent and replicable speed-threshold rules rather than a supervised model that requires labeled training data.

2.2.3 Map Matching

Map matching is a technique for aligning position data from tracking devices with a road map [1]. Based on the taxonomy of Quddus et al. (2007), *map matching* algorithms are classified into four categories:

1. **Geometric:** Uses the nearest distance between GPS points and road segments (*point-to-point*, *point-to-curve*).
2. **Topological:** Adds consideration of road-network connectivity.
3. **Probabilistic:** Uses probability models such as the *Hidden Markov Model* (HMM) [6].
4. **Advanced (advanced):** Uses *fuzzy logic*, *Kalman filter*, or combinations of methods.

The selection of a *map matching* method depends on road-network complexity and GPS data sampling frequency. Newson and Krumm (2009) showed that HMM methods are effective at sampling intervals below 30 seconds, but their accuracy decreases significantly at longer intervals [6]. For simple road networks with low topological complexity, geometric methods are sufficiently adequate [1].

2.2.4 Trajectory Segmentation

Trajectory segmentation is the process of dividing a sequence of GPS points into meaningful sub-trajectories [7]. According to Zheng (2015), segmentation can be performed based on:

1. **Time interval:** Separating trajectories when the time gap between points exceeds a specified threshold.
2. **Spatial shape:** Based on geometric properties such as turning points.
3. **Semantic meaning:** Based on activities or stops (*stay point detection*).

Time-threshold values are heuristic and depend on data characteristics and the application [7]. Nurlita (2024) used a 10-minute threshold on the same dataset, referring to the finding that observation *gaps* of more than 10 minutes significantly reduce analytical accuracy.

2.2.5 *Origin-Destination (OD) Matrix*

An OD matrix is a tabular representation of the number of trips between pairs of origin and destination zones in a transportation network. Alexander et al. (2015) showed that OD matrices extracted from mobile-phone data correlate well with household survey data, especially at coarser spatial aggregation levels [5].

2.2.6 *Home-Location Estimation and Catchment Area*

The *anchor point* model introduced by Ahas et al. (2010) identifies meaningful locations for mobile-phone users based on temporal patterns of activity [10]. Home location is estimated from the location with the highest frequency during the nighttime period (22:00–06:00), whereas workplace location is identified from activity during working hours.

2.2.7 *Traffic Density*

Traffic density is defined as the number of vehicles per unit length of road at a given time. In the MPD context, density is measured by the number of unique MAIDs detected within an intersection zone per unit time and then normalized by the number of observation days.

CHAPTER III

RESEARCH METHODOLOGY

This chapter describes the methodology used to achieve the research objectives. The overall research workflow is shown in Figure 3.1.

3.1 Research Workflow

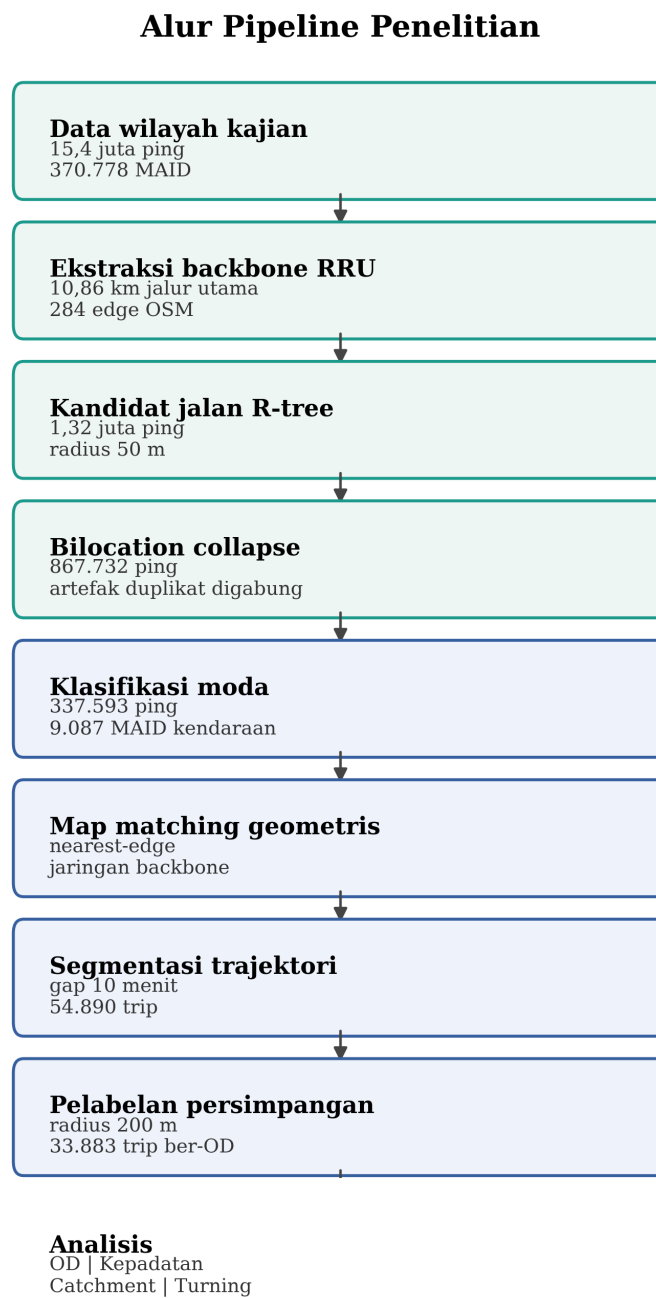


Figure 3.1. Research pipeline workflow from the study-area data to the analysis stage

3.2 Research Data

3.2.1 Data Source

The research data consist of active MPD obtained through a collaboration between the research team and a telecommunications company in Indonesia. The dataset covers GPS data from the entire Special Region of Yogyakarta for the period from October 2021 to June 2022.

3.2.2 Data Specification

The initial raw mobility dataset is `all_gps_data_no_dup.parquet`, a large active-MPD parquet file containing GPS observations before spatial extraction. The working dataset used in the research pipeline is the extracted study-area data for Ring Road Utara (`output_bbox.parquet`), comprising 15,386,869 GPS points from 370,778 unique MAIDs. Each data point contains four main attributes: `maid` (device identity), `latitude`, `longitude`, and `timestamp` (in *Unix Timestamp* format). After extraction using the QGIS-curated *fishbone* network, road-candidate selection, *bilocation collapse*, and transport-mode classification, 312,935 GPS points from 8,980 MAIDs associated with vehicle-likelihood movement on the RRU corridor were obtained. The same QGIS-curated *backbone* network was then used for the OD and *catchment area* analyses; under the current network revision, this retained 312,935 GPS points from 8,980 MAIDs before trajectory segmentation.

3.2.3 Road Network

The Ring Road Utara road network was modeled in two complementary forms. First, the *fishbone* network was used as the main preprocessing network because it includes the west–east RRU main route together with north–south branches around six intersections. Second, the *backbone-only* network was extracted from this network for the OD and *catchment area* analyses, ensuring that the results of these two analyses represent only vehicles that actually traversed the RRU main route. The six intersection points used as analysis zones are Kronggahan, Jombor, Monjali, Kentungan, Condongcatur, and UPN.

The road network was constructed through the following stages:

1. Downloading or loading the OSM road graph for the Sleman area using the vehicle (*drive*) network type.
2. Selecting *trunk/primary* road segments with names corresponding to the RRU main route, such as Jl. Siliwangi, Ring Road Utara, Padjajaran/Pajajaran, and Bunderan Jombor.

3. Determining the westernmost node, the waypoint node at Bunderan Jombor, and the easternmost node, and then constructing the shortest length-weighted path for the west–Jombor–east route. The Jombor waypoint was used so that the path follows the roundabout rather than the parallel bypass segment.
4. The OSM-derived network was reviewed and cleaned manually in QGIS. The authoritative *fishbone* input used in the final pipeline is `data/processed/network/rru_with_`.
5. The authoritative analytical *backbone* input is `data/processed/network/rru_backbone_`. OSM-generated files under `data/external/generated_network/` are retained only as reference artifacts and are not used as the final analytical network.

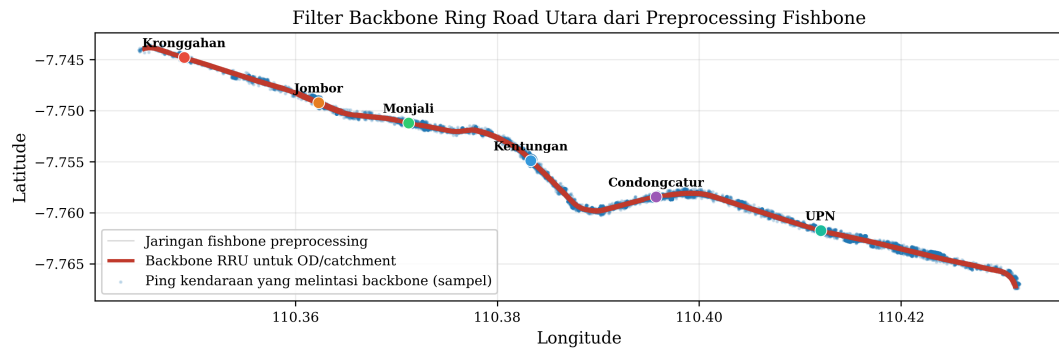


Figure 3.2. Preprocessing *fishbone* network, Ring Road Utara *backbone*, and sample GPS points of vehicles passing the *backbone* filter

3.3 Data Preprocessing

Data preprocessing was carried out in several stages to filter the raw data into an analysis-ready dataset. A summary of data reduction at each stage is shown in Table 3.1.

Table 3.1. Data Reduction at Each Preprocessing Stage

Stage	GPS Points	Unique MAIDs	Percentage
Study-area data (<code>output_bbox</code>)	15,386,869	370,778	100.0%
<i>R-tree candidates on the fishbone</i>	1,091,423	52,028	7.1%
<i>Bilocation collapse</i>	708,975	48,395	4.6%
Motorized-vehicle-likelihood classification	312,935	8,980	2.0%
Analytical <i>backbone</i> filter	312,935	8,980	2.0%

3.3.1 Bounding Box Filter

The first stage in the working pipeline used data already contained within the Ring Road Utara *bounding box*. These data served as the initial input for road-candidate filtering; therefore, all reduction percentages in Table 3.1 were calculated relative to `output_bbox.parquet`.

3.3.2 *R-tree Edge Candidates*

Using a spatial *R-tree* index, each GPS point was paired with candidate road segments on the *fishbone* network within a 50-meter radius. Points without candidates were eliminated, so the initial preprocessing retained vehicles interacting with the RRU corridor and its branches around the intersections.

3.3.3 *Bilocation Collapse*

This stage merged duplicate points appearing at the same time (*bilocation artifacts*) as a result of the GPS sampling mechanism.

3.3.4 Transport-Mode Classification

After the *bilocation collapse* stage, speed-feature-based transport-mode classification was performed to retain MAIDs indicating motorized-vehicle movement. This approach follows the literature on GPS-based mode classification, which uses speed features such as maximum speed, average speed, and speed percentiles to distinguish walking, cycling, and motorized-vehicle modes [8, 9]. In this study, a MAID was classified as a motorized vehicle if its maximum inter-ping speed reached at least 25 km/h or its 95th-percentile speed reached at least 15 km/h. These thresholds were used conservatively to remove stationary users, pedestrians, and bicycle-like patterns while retaining vehicles that were observed mostly under slow-moving or congested conditions. This classification is also consistent with the context of vehicle and non-vehicle classification in intersection GPS data in the study by Rakhman et al. [4].

3.4 Map Matching

Map matching was performed using the geometric *nearest-edge* method (point-to-curve matching) based on the taxonomy of Quddus et al. (2007) [1]. For each GPS point, the nearest candidate road segment was selected based on the Euclidean distance among candidates computed by the *R-tree* index.

The selection of a geometric method instead of a *Hidden Markov Model* (HMM) was based on two considerations:

1. **Low network complexity:** The *fishbone* network consists only of one west–east main corridor and limited branches around intersections, so topological ambiguity remains low.
2. **Low sampling frequency:** Newson and Krumm (2009) showed that HMM accuracy decreases significantly at sampling intervals above 30 seconds [6]. The data in this study have inter-ping *gaps* ranging from minutes to hours.

3.4.1 Analytical *Backbone* Filter

After vehicle-likelihood MAIDs were obtained from *fishbone* preprocessing, their GPS points were reprocessed against the QGIS-curated *backbone* network. This stage used the same R-tree index and *nearest-edge* method, but with the authoritative *backbone* GeoJSON. Points having a candidate *backbone* edge within a 50-meter radius were retained as the OD and *catchment area* analysis dataset. Thus, the OD analysis for the six intersections and the *catchment area* analysis were constructed only from vehicles traversing the RRU *backbone*, whereas density and *turning movement* analyses continued to use intersection information from the *fishbone* network.

3.5 Trajectory Segmentation

Each sequence of pings for a MAID was segmented into travel units (*trips*) using a time-based segmentation method [7]. The threshold used was 10 minutes (600 seconds), following Nurlita (2024), who used the same threshold on a similar dataset.

Criteria for a valid *trip*:

- At least 2 GPS points per *trip*
- A *trip* with an identified origin–destination pair must have at least 1 point within an intersection zone

3.6 Intersection Labeling

Each map-matched GPS point was labeled based on its proximity to an intersection. An intersection zone was defined as a circle with a radius of 200 meters from the intersection center point.

For each *trip*, the following were determined:

- **Origin:** The first intersection detected in the *trip* sequence
- **Destination:** The last intersection detected in the *trip* sequence

3.7 Analysis Modules

3.7.1 Origin-Destination Matrix

A 6×6 OD matrix was constructed from the *origin-destination* pair of each *trip* in the *backbone* dataset. The matrix was computed for several temporal slices:

- Overall aggregate
- Hourly (24 matrices)
- Weekdays versus weekends

- Peak hours (07:00–09:00 and 16:00–18:00 WIB) versus off-peak hours
- Monthly (9 matrices, from October 2021 to June 2022)

3.7.2 Traffic Density Estimation

Traffic density was estimated from the number of unique MAIDs detected in intersection zones per time window. The calculation was performed at 15-minute and 1-hour resolutions and then normalized by the number of observation days.

3.7.3 Catchment Area Mapping

Users' home locations were estimated using an *anchor point* model [10]:

1. For each MAID appearing in the Ring Road Utara *backbone* dataset, all ping data were extracted from the broader *bounding box* dataset (15.4 million pings).
2. The home location was estimated as the grid cell with the highest ping frequency during the nighttime period (22:00–06:00 WIB).
3. If no nighttime data were available, the location with the highest overall frequency was used.
4. Home locations were aggregated into a spatial grid measuring $\sim 550 \text{ m} \times 550 \text{ m}$.

3.7.4 Turning Movement Classification

At each intersection, movement patterns were classified from the *fishbone* dataset based on the positions of the *trip* entry and exit points relative to the *backbone*:

- **Through:** Entering and exiting from different *backbone* segments
- **U-turn:** The *origin* and *destination* are at the same intersection

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results of all analytical stages conducted in this study, beginning with data preprocessing, *map matching*, and trajectory segmentation, followed by analyses of OD matrices, traffic density, *catchment area*, and *turning movement*.

4.1 Preprocessing and *Map Matching* Results

4.1.1 Summary of *Map Matching*

The *map matching* process was carried out in two stages. The first stage mapped 312,935 GPS points from vehicle-likelihood MAIDs to the RRU *fishbone* network in order to preserve intersection context. The second stage filtered the dataset to the RRU *backbone* network, yielding 312,935 GPS points from vehicle-likelihood MAIDs retained by the QGIS-curated backbone network. The statistics in Table 4.1 summarize the matching results for the *backbone* dataset used in the OD and *catchment area* analyses. The matching-distance statistics are shown in Table 4.1.

Table 4.1. *Map Matching* Distance Statistics

Metric	Value
Matched GPS points	312,935
Unique MAIDs	8,980
Unique road segments	593
Mean matching distance	11.7 m
Median distance (P50)	6.9 m
90th percentile (P90)	33.1 m
99th percentile (P99)	39.5 m

The median matching distance of 6.9 m indicates that most GPS points are already located close to the road network, supporting the use of a geometric matching method.

4.1.2 Trajectory Segmentation Results

Segmenting the *backbone* dataset with a 10-minute time-gap threshold produced 50,791 travel units (*trips*) from 8,903 MAIDs. The *trip* statistics are shown in Table 4.2.

The low median number of pings per *trip* (3 pings) confirms the low sampling-frequency characteristics of the dataset.

Table 4.2. Trajectory Segmentation Statistics

Metric	Value
Total <i>trips</i>	50,791
Total pings (in valid <i>trips</i>)	272,906
MAIDs with valid <i>trips</i>	8,903
Mean <i>trips</i> /MAID	5.7
Median <i>trip</i> duration	217 seconds
Mean <i>trip</i> duration	348 seconds
Median pings/ <i>trip</i>	3

4.1.3 Intersection Zone Distribution

Of the 272,906 pings in valid *trips* in the current *backbone* dataset, all pings were labeled to one of the six intersection zones because the current QGIS-clean network is closely aligned with the intersection-zone corridor representation. The distribution by intersection is shown in Table 4.3.

Table 4.3. Ping Distribution within Intersection Zones

Intersection	Number of Pings	Unique MAIDs
UPN	88,350	5,666
Condongcatur	68,283	6,324
Monjali	41,328	5,463
Kentungan	38,236	5,408
Jombor	22,192	4,153
Kronggahan	14,517	2,709
Total	272,906	–

4.2 Origin-Destination Matrix Analysis

4.2.1 Aggregate OD Matrix

Of the 50,791 *trips* in the *backbone* dataset, 33,288 *trips* had identifiable OD pairs. The aggregate 6×6 OD matrix is shown in Table 4.4.

Several important findings emerge from the OD matrix:

1. **Diagonal dominance:** The largest values are located on the main diagonal (O=D), indicating many short *trips* that were detected in only one intersection zone. This pattern is an artefact of the low sampling frequency.
2. **Busiest intersections:** UPN and Condongcatur have the largest total flows based on row/column origin-destination totals, while the largest diagonal values occur at

Table 4.4. Aggregate OD Matrix (October 2021–June 2022)

O \ D	Kron.	Jomb.	Monj.	Kent.	Cond.	UPN
Kronggahan	1,518	393	430	143	299	215
Jombor	408	1,849	715	257	391	279
Monjali	403	518	3,893	537	817	556
Kentungan	184	184	664	3,313	612	388
Condongcatur	266	243	903	553	4,676	1,077
UPN	161	148	537	230	752	4,776

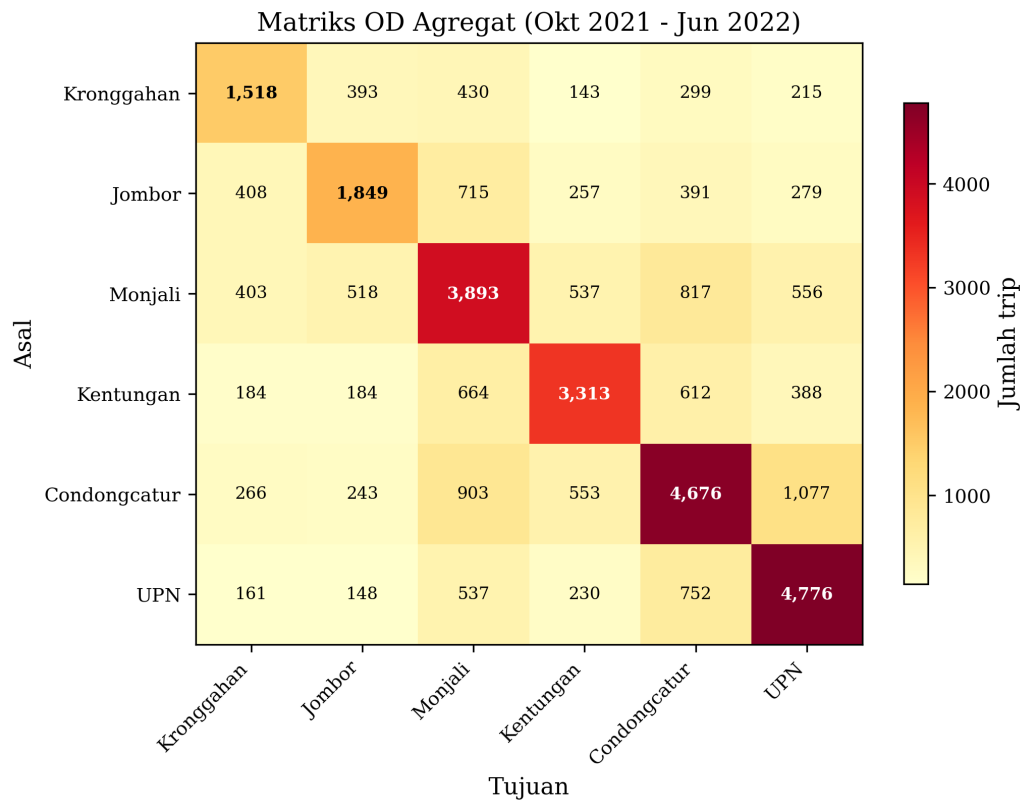


Figure 4.3. Visualization of the aggregate OD matrix for the six Ring Road Utara intersections

UPN (4,776), Condongcatur (4,676), and Monjali (3,893).

3. **Dominant non-diagonal corridor:** The largest inter-intersection flow occurs between Condongcatur and UPN. The two-way total for this pair reaches 1,829 *trips*, consisting of 1,077 *trips* from Condongcatur to UPN and 752 *trips* from UPN to Condongcatur.

4.2.2 Weekday and Weekend Comparison

The OD matrices show a substantial difference between weekdays and weekends:

- **Weekdays:** 23,005 *trips* (69.1% of the total)

- **Weekends:** 10,283 *trips* (30.9% of the total)

The weekday-to-weekend ratio ($\sim 2.3:1$) is consistent with a commuter pattern dominated by work and educational activities on weekdays.

4.3 Traffic Density Analysis

4.3.1 Hourly Density Profile

The hourly density profile shows a clear pattern across all intersections for vehicles observed on the preprocessed *fishbone* network. Condongcatur and Monjali are the two zones with the highest density. The general observed pattern is as follows:

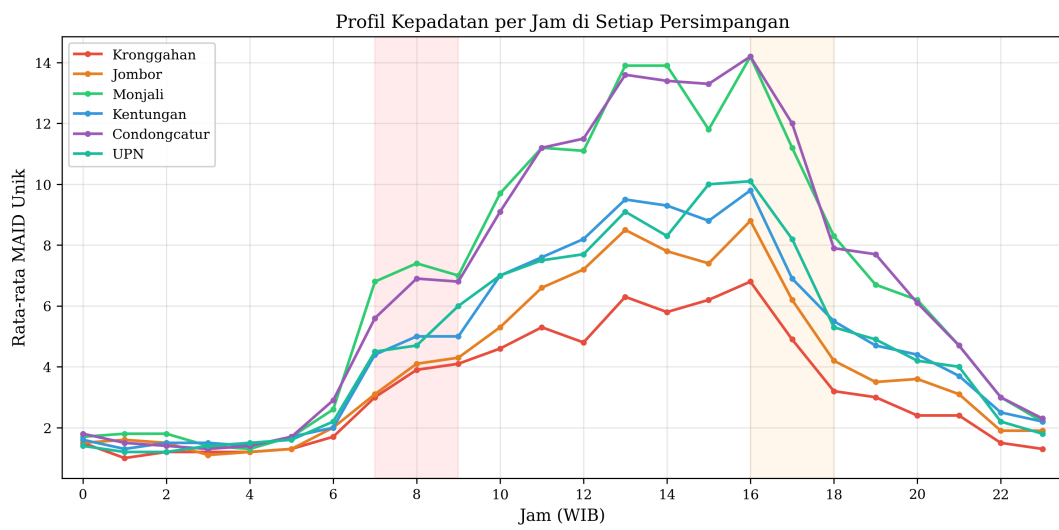


Figure 4.4. Hourly traffic-density profile based on average unique MAIDs

- **Morning busy period (07:00–09:00 WIB):** A gradual increase in density
- **Midday peak period (11:00–14:00 WIB):** The highest density at most intersections
- **Afternoon busy period (16:00–18:00 WIB):** A second peak, particularly pronounced at Condongcatur
- **Nighttime (22:00–06:00 WIB):** A sharp decline, averaging 1–2 MAIDs per hour

4.3.2 Intersection Density Ranking

Table 4.5 presents the ranking of average daily density by intersection.

Condongcatur ranks first because this intersection connects Ring Road Utara with Jl. Affandi (a main access route to UGM) and Jl. Laksda Adisucipto (a national road).

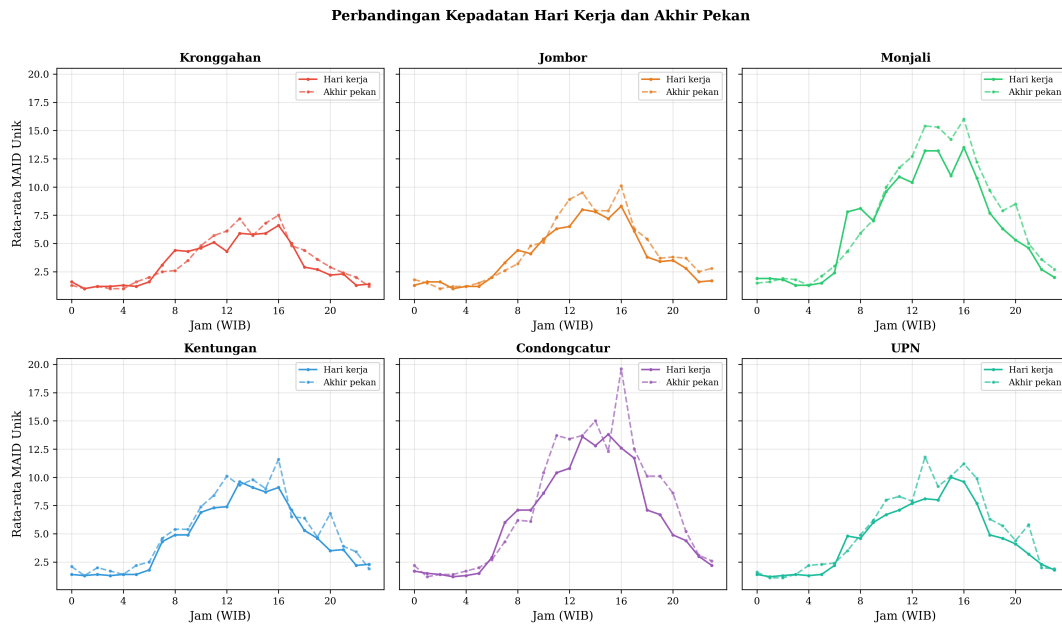


Figure 4.5. Comparison of weekday and weekend density profiles

Table 4.5. Average Daily Density Ranking

Rank	Intersection	Daily Average	Total Unique MAIDs
1	Condongcatur	49.4	6,324
2	Monjali	46.7	5,463
3	Kentungan	37.0	5,408
4	UPN	35.4	5,666
5	Jombor	28.7	4,153
6	Kronggahan	22.6	2,709

4.4 Catchment Area Analysis

Home-location estimation was successfully performed for all 8,903 vehicle-likelihood MAIDs with valid *trips* on the Ring Road Utara *backbone* dataset. The results show that:

1. **Spatial concentration:** The highest-ranking grid cells for all intersections are concentrated around Ring Road Utara itself (coordinates $\sim 7.75^{\circ}\text{S}$, $110.37\text{--}110.41^{\circ}\text{E}$), indicating that most RRU users live near the corridor.
2. **Variation across intersections:** Condongcatur and UPN show a more concentrated *catchment* distribution on the eastern side, consistent with their proximity to residential areas in Condongcatur and Depok.
3. **Number of grid cells:** Each intersection contains 110–111 occupied grid cells, indicating that RRU users originate from a relatively broad area despite being concentrated near the corridor.

Sebaran Catchment Area Berdasarkan Estimasi Lokasi Rumah

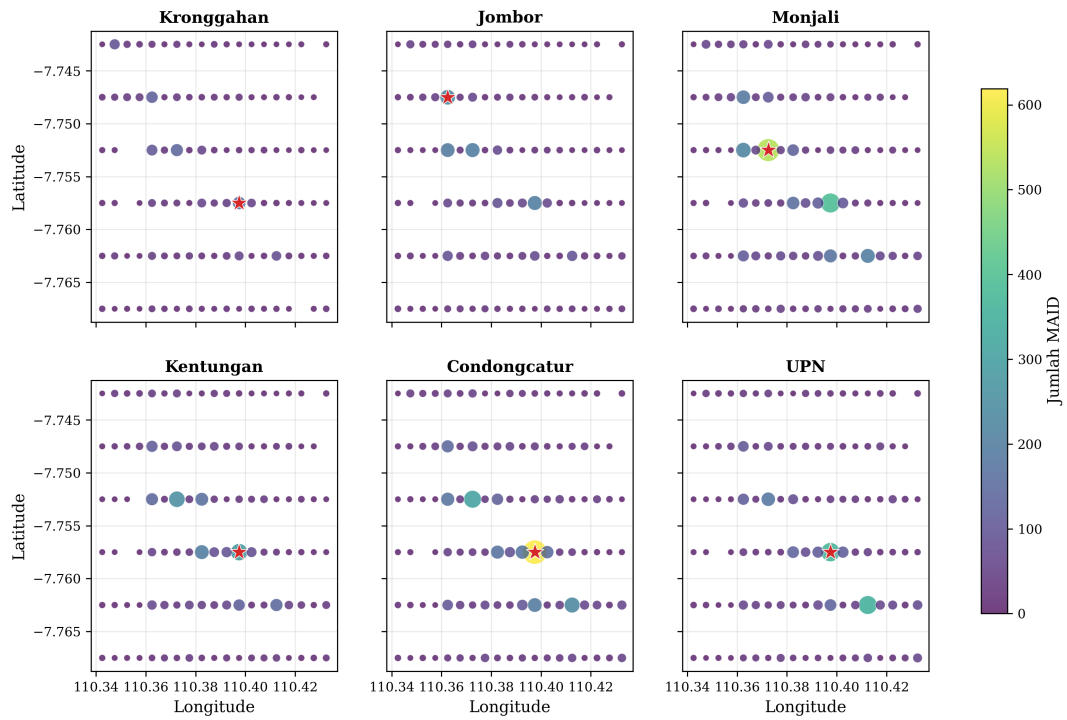


Figure 4.6. Distribution of the *catchment area* based on estimated home locations of Ring Road Utara users

4.5 Turning Movement Analysis

The *turning movement* classification on the *fishbone* dataset identified 26 movement patterns across all intersections. The main findings are as follows:

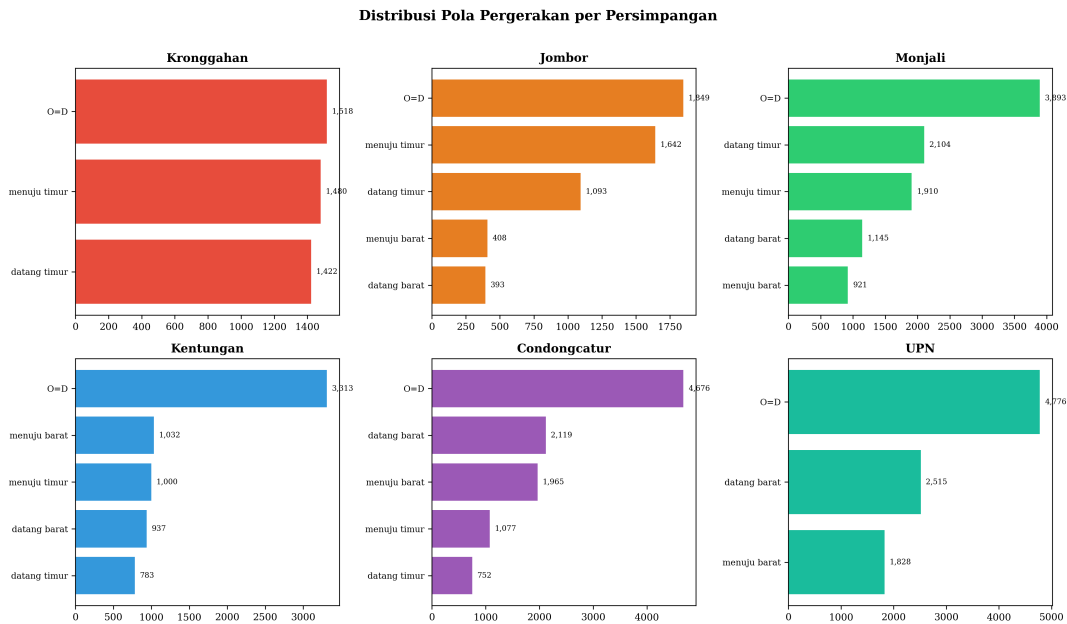


Figure 4.7. Distribution of movement patterns from OD classification by intersection

1. **Dominance of *U-turns*:** The *u-turn* pattern (O=D) appears as the largest category at all intersections, which is a consequence of the low sampling frequency (many *trips* are observed at only one intersection).
2. **Arriving and departing patterns:** After *u-turns*, the *arriving from west* and *departing west* patterns are the most dominant, consistent with the role of Ring Road Utara as an east-west corridor.
3. **Condongcatur:** This intersection exhibits the most complete variation in *turning movement*, with 5 identified categories, including substantial east-west movements.

4.6 Discussion

4.6.1 Validity of the Results

The findings of this study are consistent with several findings from previous studies:

- Diagonal dominance in the OD matrix is a common finding in MPD-based studies with low sampling frequency, as reported by Alexander et al. (2015) [5].
- The density pattern shows an increase during midday to afternoon activity periods. This finding supports the presence of a daily temporal traffic structure, although it does not fully follow the initial assumption of a dominant morning peak.
- Condongcatur as the busiest intersection is consistent with hypothesis H1, which predicts high volume at intersections connected to major arterial roads.

4.6.2 Limitations

1. **Low sampling frequency:** The median of 3 pings per *trip* limits the ability to identify *turning movement* accurately. Many *trips* are observed at only one intersection, producing artificial *u-turns* in the OD matrix.
2. **Sample bias:** Active MPD only covers *smartphone* users whose applications request location permission, and therefore does not represent the entire road-user population.
3. **Absence of ground truth:** Manual traffic survey data are not available for direct validation of the estimated results. Validation can therefore only be conducted qualitatively against expected patterns.
4. **COVID-19 effects:** The data period (October 2021–June 2022) includes the PPKM restriction period, which may have affected normal mobility patterns.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This study characterized the pre-toll traffic *baseline* along the Ring Road Utara corridor in Yogyakarta using large-scale active *Mobile Positioning Data* (MPD). The data were preprocessed on the *fishbone* network and filtered on the RRU *backbone* network for OD and *catchment area* analyses. The conclusions of this study are as follows:

1. **Origin-Destination matrix:** A 6×6 OD matrix was successfully extracted from 33,288 identified *trips*. Condongcatur was the intersection with the highest total flow, and the Condongcatur–UPN corridor showed the strongest connectivity. Week-day flows dominated (69.1%) compared with weekend flows (30.9%), indicating a strong commuter pattern.
2. **Traffic density:** Spatiotemporal density profiles were identified across all 6 intersections. Condongcatur had the highest density ranking (average 49.4 MAIDs/day), followed by Monjali (46.7) and Kentungan (37.0). Density peaks at most intersections occurred from midday to the afternoon, particularly around 13:00 and 16:00 WIB, whereas the morning period showed a more gradual increase.
3. **Catchment area:** Home-location estimation using the *anchor point* model was successfully performed for all 8,903 vehicle-likelihood MAIDs with valid *trips* on the RRU *backbone*. Most Ring Road Utara users live in areas near the corridor, with the highest concentration in grid cells around Condongcatur and its surrounding areas.
4. **Turning movement:** Turning-movement classification identified 26 patterns across the intersections. However, the largest category at every intersection was artificial *u-turns* (O=D), primarily reflecting the large number of *trips* observed in only one intersection zone because of the low sampling frequency. Accordingly, the *turning movement* results are more appropriately positioned as exploratory analysis, while the *arriving* and *departing* patterns are used as more cautious indicators of movement direction.

Overall, this study demonstrates that active MPD can be used to construct a comprehensive intersection-level traffic *baseline*, despite limitations related to the data sampling frequency.

5.2 Recommendations

Based on the findings and limitations of this study, several recommendations for future research are proposed:

1. **Post-toll impact study:** Conduct a similar analysis after the toll road begins operation and compare it with the *baseline* data from this study to measure the actual impact on traffic patterns along Ring Road Utara.
2. **Improved trajectory reconstruction:** Implement *path reconstruction* or *map matching* methods that incorporate network-topology constraints to improve accuracy for *trips* with very few pings. HMM methods may be tested for comparison, but they should be accompanied by parameter calibration and validation because the dataset in this study has low sampling intervals.
3. **Validation using survey data:** Conduct quantitative validation by comparing the estimated density results with manual traffic-survey data from the Yogyakarta Special Region Transportation Agency (Dinas Perhubungan DIY).
4. **Integration of additional data:** Combine MPD with other data sources, such as weather data, event calendars, and PPKM data, to enable more in-depth multivariate analysis.
5. **Parameter sensitivity analysis:** Conduct sensitivity analysis on key parameters, such as segmentation thresholds, intersection-zone radius, and speed thresholds, to evaluate the robustness of the results.

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APPENDIX

L.1 Appendix Contents

The appendix is optional and depends on the agreement with the thesis supervisor. It may include:

1. Evidence of questionnaire implementation, such as questionnaire items, summaries of respondent answers, and questionnaire documentation.
2. Specifications of the application or system developed, including technical specifications, application download links, user manuals, and application screenshots.
3. Important code excerpts that need to be included.
4. Tables that are too long to present in the main body of the thesis but remain necessary for documentation or analysis.
5. Supporting figures that are not essential for the main body but support the argument, observation, or analysis.
6. Derivations of formulas or proofs of theorems that are too long and technical for the main text, allowing advanced readers to examine the derivation process in greater detail.